

Md Mahadi Hassan

Ph.D. Candidate, Department of Computer Science, University of Central Florida

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Research Profile

My research studies **evaluation integrity** by asking when apparent success in AI systems reflects genuine capability. I focus on cases where benchmark scores are inflated by prior exposure or generated solutions appear correct but fail to achieve their intended outcomes. I develop contamination-resistant private evaluation protocols, empirical benchmark audits, and execution-grounded evaluation methods that reveal when scores are distorted by hidden exposure, benchmark-specific optimization, or inadequate behavioral proxies. Across natural language processing and software engineering, I use **state reconciliation** as a conceptual framework for identifying mismatches between an evaluation's assumptions and the model behavior or system outcomes that can actually be observed.

Research Interests: Machine Learning Evaluation and Benchmarking; Applied Natural Language Processing; AI for Software Engineering.

Education

University of Central Florida (UCF), Orlando, Florida, USA

Fall 2024 – Present

Ph.D. in Computer Science, **expected Summer 2027**

Advisor: Prof. Shubhra Kanti Karmaker (Santu)

Dissertation research: Evaluation integrity in language-model benchmarks, with emphasis on contamination, hidden exposure, and private black-box evaluation.

Auburn University, Auburn, Alabama, USA

Fall 2021 – Summer 2024

M.S. in Computer Science and Software Engineering

Advisor: Prof. Shubhra Kanti Karmaker (Santu)

Bangladesh University of Engineering and Technology (BUET), Dhaka, *2012 – 2017*
Bangladesh

B.Sc. in Computer Science and Engineering

Publications

Peer-Reviewed Publications

1. **Md. Mahadi Hassan**, Sazzadur Rahaman, Yusuf Sarwar Uddin, and Santu Karmaker. “**Private Evaluation Protocol: Contamination-Resistant Black-Box Evaluation via Public-Slice Surrogates.**” Proceedings of the AACL-IJCNLP 2026.
2. **Md. Mahadi Hassan**, John Salvador, Akond Ashfaqur Rahman, and Shubhra Kanti Karmaker Santu. “**Large Language Models for IT Automation Tasks: Are We There Yet?**” *Findings of the Association for Computational Linguistics: ACL 2026.*

3. Eshgin Hasanov, **Md Mahadi Hassan**, Santu Karmaker, and Aashish Yadavally. **“The Path Not Taken: Duality in Reasoning about Program Execution.”** *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics: ACL 2026*.
4. Eftekhari Hossain, Sanjeev Kumar Sinha, Naman Bansal, Alex Knipper, Souvika Sarkar, John Salvador, Yash Mahajan, Sri Guttikonda, Mousumi Akter, **Md. Mahadi Hassan**, Matthew Freestone, Matthew C. Williams Jr., Dongji Feng, and Santu Karmaker. **“LLMs as Meta-Reviewers’ Assistants: A Case Study.”** *Proceedings of the 2025 Conference of the Americas Chapter of the Association for Computational Linguistics: NAACL 2025*.
5. **Md Mahadi Hassan**, John Salvador, Shubhra Kanti Karmaker Santu, and Akond Rahman. **“State Reconciliation Defects in Infrastructure as Code.”** *Proceedings of the ACM on Software Engineering, Volume 1, FSE, Article 83, 2024*.
6. **Md Mahadi Hassan**, Alex Knipper, and Shubhra Kanti Karmaker Santu. **“Introducing ‘Forecast Utterance’ for Conversational Data Science.”** *Transactions on Machine Learning Research (TMLR), 2024*.
7. Souvika Sarkar, Mohammad Fakhruddin Babar, **Md Mahadi Hassan**, Monowar Hasan, and Shubhra Kanti Karmaker Santu. **“Processing Natural Language on Embedded Devices: How Well Do Transformer Models Perform?”** *Proceedings of the 15th ACM/SPEC International Conference on Performance Engineering (ICPE 2024), pp. 211–222, 2024*.
8. R. Alexander Knipper, **Md. Mahadi Hassan**, Mehdi Sadi, and Shubhra Kanti Karmaker Santu. **“Analogy-Guided Evolutionary Pretraining of Binary Word Embeddings.”** *Proceedings of AACL-IJCNLP 2022, pp. 683–693, 2022*.
9. Shubhra Kanti Karmaker Santu, **Md. Mahadi Hassan**, Micah J. Smith, Lei Xu, ChengXiang Zhai, and Kalyan Veeramachaneni. **“AutoML to Date and Beyond: Challenges and Opportunities.”** *ACM Computing Surveys, 54(8), Article 175, 2021*.

Manuscripts Under Review

1. Eshgin Hasanov, **Md. Mahadi Hassan**, Santu Karmaker, and Aashish Yadavally. **“Beyond Exact Match: The Spectrum of Repetitiveness in Code.”** Manuscript under review.

Preprints and Technical Reports

1. Yogesh Agrawal, Aniruddha Dutta, **Md Mahadi Hassan**, Santu Karmaker, and Aritra Dutta. **“FinTradeBench: A Financial Reasoning Benchmark for LLMs.”** *arXiv:2603.19225, 2026*.
2. **Md Mahadi Hassan**, Alex Knipper, and Shubhra Kanti Karmaker Santu. **“Large Language Models as Your Personal Data Scientist.”** *arXiv:2305.13657, 2023*.

Selected Research Artifacts and Evaluation Infrastructure

- **Private Evaluation Protocol (PEP):** A controlled-exposure framework for contamination-resistant black-box evaluation using public-slice surrogates while preserving the private benchmark. Associated with AACL-IJCNLP 2026.
- **ExITBench:** An execution-based benchmark containing 126 real-world IT-automation tasks and 733 test cases in controlled environments; evaluates whether generated automation reaches the intended system state. Associated with [ACL Findings 2026](#).
- **DEXBENCH:** A benchmark of 445 paired instances for jointly evaluating forward execution reasoning and backward counterfactual reasoning in code models. [Code and data](#).

- **State-Reconciliation Defect Dataset and Taxonomy:** An empirical dataset of 5,110 Infrastructure-as-Code defects and an eight-category taxonomy used to derive LLM-based defect-detection heuristics. Associated with [FSE 2024](#).

Sponsored Research and Proposal Development

Office of Naval Research (ONR) – *Rapid Holistic Summarization for Decision Superiority in Urgent Scenarios*

PI: Prof. Shubhra Kanti Karmaker Santu

Role: Graduate research contributor. Developed the proposal’s central motivation figure, synthesized relevant prior work, and contributed to the design and presentation of the evaluation framework.

Status: Funded

National Science Foundation (NSF), STEM for All – *ViDS: The Gateway to STEM K-12 Data Science Training and Education with Pedagogical AI Agents*

PIs: Prof. Shubhra Kanti Karmaker Santu and Prof. Roger Azevedo

Role: Lead graduate researcher for proposal development. Drafted the proposal from the initial concept through submission; formulated the research thrusts; co-developed the evaluation and education plans with the PIs; and created the proposal’s figures, workflow diagrams, project timeline, and milestone plan.

Status: Under review

Awards and Research Support

- **NSF ACCESS Computing Allocation**, 2026 – 2027. Principal Investigator; **200,000** ACCESS credits for GPU computing.
- **Woltosz Fellowship**, Auburn University, 2021–2024 – competitive graduate fellowship.

Research Experience

Graduate Research and Teaching Assistant, University of Central Florida *August 2024 – Present*

Advisor: Prof. Shubhra Kanti Karmaker Santu, Bridge-AI Lab, Orlando, FL

- Lead research on evaluation integrity for language models, with emphasis on hidden benchmark exposure, contamination, benchmark-specific optimization, and trustworthy black-box measurement.
- Developed the **Private Evaluation Protocol**, a public-slice surrogate framework for contamination-resistant private evaluation; designed controlled-exposure experiments to distinguish exposure advantage from transferable capability; the work was accepted at AACL-IJCNLP 2026.
- Led the development of **ExITBench**, including execution-based evaluation environments and failure analysis for LLM-generated IT automation; the work resulted in **ACL Findings 2026**.

Graduate Research and Teaching Assistant, Auburn University *August 2021 – August 2024*

Advisor: Prof. Shubhra Kanti Karmaker Santu, Big Data Intelligence Lab, Auburn, AL

- Conducted an empirical study of 5,110 Infrastructure-as-Code defects, developed a taxonomy of state-reconciliation failures, and translated the taxonomy into LLM-based defect-detection heuristics (**FSE 2024**).

- Introduced the **Forecast Utterance** formulation and developed zero-shot methods for translating natural-language forecasting requests into structured machine-learning tasks (**TMLR 2024**).
- Contributed to a taxonomy and research roadmap for end-to-end automated machine learning (**ACM Computing Surveys 2022**).
- Collaborated on efficient NLP research, including binary word embeddings and empirical deployment studies of transformer models on resource-constrained devices (**AACL-IJCNLP 2022**; **ICPE 2024**).

Teaching Experience

Graduate Teaching Assistant, University of Central Florida *Fall 2026 — Spring 2027*

Department of Computer Science, Orlando, Florida, USA

Courses:

- COP 3502: Computer Science I *Fall 2026; Spring 2027*

Graduate Teaching Assistant, Auburn University *Fall 2021 — Spring 2024*

Department of Computer Science and Software Engineering, Auburn, Alabama, USA

Courses:

- CPSC 5120: Database I *Fall 2021; Spring 2022; Spring 2023; Spring 2024*
- CPSC 5210: Web Application Development with JSP *Fall 2023; Spring 2024*
- CPSC 3280: Algorithms II *Fall 2021; Spring 2022*
- CPSC 3300: Computer Organization and Assembly Language Programming *Spring 2022*
- CPSC 3270: Introduction to Algorithms *Fall 2021*

Selected Research Presentations

2026	Annual Meeting of the Association for Computational Linguistics (ACL)	<i>Large Language Models for IT Automation Tasks: Are We There Yet?</i>	Poster
2026	UCF AI Day, University of Central Florida	<i>Private Evaluation Protocol: Contamination-Resistant Black-Box Evaluation</i>	Poster
2024	SouthNLP	<i>LLM as Your Personal Data Scientist</i>	Poster
2023	Auburn University Research Symposium, Huntsville, Alabama	<i>Introducing Forecast Utterance for Conversational Data Science</i>	Poster
2022	Auburn University Research Symposium, Auburn University	<i>Introducing Conversational Data Science</i>	Poster

Professional Service

Reviewer

- ACL Rolling Review (ARR), 2023, 2025, 2026.
- NeurIPS, 2026.
- Transactions on Machine Learning Research (TMLR), 2024, 2025.
- EMNLP, 2023.
- EMNLP Workshop on Bangla Language Processing, 2023.

Program and Track Committees

- ACL, Track Committee Member, 2023.
- EMNLP, Track Committee Member, 2022.

Conference Service

- **ACL 2026:** Conference Volunteer.

Professional Memberships

- Association for Computational Linguistics (ACL), 2026–Present

Industry Experience

Senior Software Engineer, REVE Systems

September 2017 – December 2020

Purbachol, Dhaka, Bangladesh

- Designed and developed scalable communication infrastructure systems.
- Led engineering teams working on reliability, automation, and deployment-pipeline improvements.

Software Engineer I, Enosis Solutions

March 2017 – September 2017

Gulshan, Dhaka, Bangladesh

- Developed backend software solutions and data-driven tooling.

Research and Technical Expertise

- **AI Evaluation and Benchmarking:** LLM evaluation, benchmark construction, controlled-exposure experiments, contamination auditing, robustness analysis, execution-grounded evaluation, and failure analysis.
- **Experimental Methodology:** Controlled experimentation, execution-based testing, qualitative coding, error-taxonomy development, and empirical software analysis.
- **Machine Learning and NLP:** PyTorch, Hugging Face Transformers, scikit-learn, parameter-efficient fine-tuning, prompt-based evaluation, and LLM inference.
- **Systems and Programming:** Python, C++, Java, Bash, Docker, Linux, Ansible, Git, GPU-accelerated experimentation, and L^AT_EX.

References

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